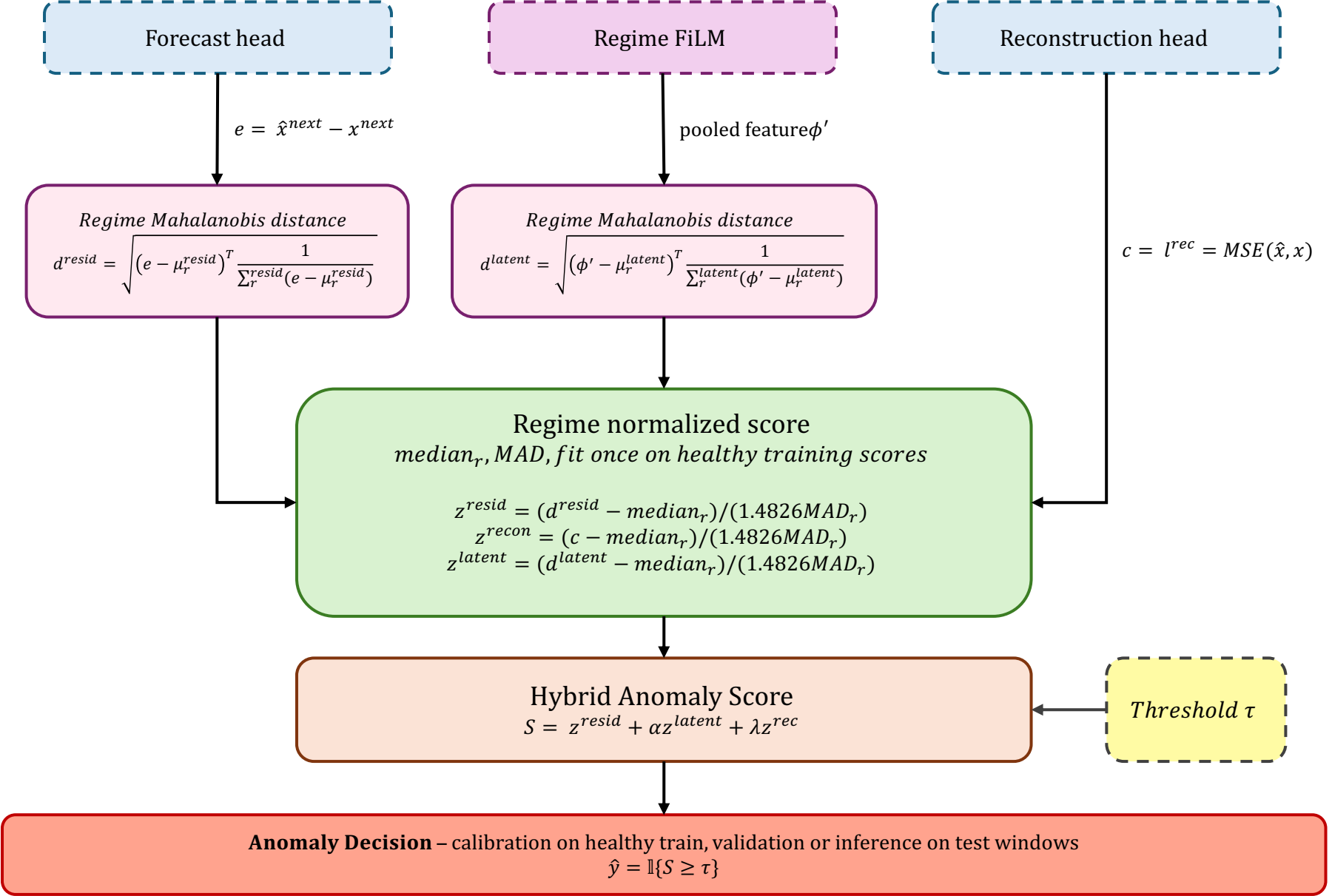
Forecast head
Regime FiLM
Reconstruction head
$e = \hat{x}^{next} - x^{next}$
pooled featureϕ'
$c = l^{rec} = MSE(\hat{x}, x)$
Regime Mahalanobis distance
$d^{resid} = \sqrt{(e - \mu_r^{resid})^T \frac{1}{\sum_r^{resid}(e - \mu_r^{resid})}}$
Regime Mahalanobis distance
$d^{latent} = \sqrt{(\phi' - \mu_r^{latent})^T \frac{1}{\sum_r^{latent}(\phi' - \mu_r^{latent})}}$
Regime normalized score
$median_r, MAD, fit\ once\ on\ healthy\ training\ scores$
$z^{resid} = (d^{resid} - median_r)/(1.4826MAD_r)$
$z^{recon} = (c - median_r)/(1.4826MAD_r)$
$z^{latent} = (d^{latent} - median_r)/(1.4826MAD_r)$
Hybrid Anomaly Score
$S = z^{resid} + \alpha z^{latent} + \lambda z^{rec}$
Threshold τ
Anomaly Decision – calibration on healthy train, validation or inference on test windows
$\hat{y} = \mathbb{I}\{S \geq \tau\}$